Group Representation Theory Notes

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Contents

1 Basics of Representation theory **2**
1.1 Morphisms of representations 3
1.2 Subrepresentations 5
1.3 Characters 10

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§1 Basics of Representation theory

Definition 1.1 (Group representation)**.** A *representation* of a group G is a morphism $\rho : G \to \mathrm{GL}(V)$, where V is a certain vector space. Commonly one refers to V as being the representation of G, taking as granted from the context the mosphism ρ.

Analogously a representation of a ring or a field is a morphism from it to an arbitrary $\mathrm{GL}(V)$. As it's usual with actions, if ρ is clear from the context we write

$$\rho_V(g)v = g \circ v = gv$$

with a slight abuse o notation.

Definition 1.2 (Group algebra)**.** Given a field k and a group G the *group algebra* of G is the set

$$kG := \left\{ \sum_{g \in G} \lambda_g g : \lambda_g \in k \right\}$$

equipped with the obvious sum (as a vector space) and the distributive multiplication that satisfies the multiplication of G.

We should prove that the one above is actually an algebra, but it is quite easy indeed. Given a representation ρ of G we can extend it linearly to a representation $\tilde{\rho}$ such that

$$\tilde{\rho} : kG \to \mathrm{End}(V)$$

and this is a ring representation (we go in End because multiplication in kG is not necessarily invertible).

Definition 1.3 (Algebra)**.** Given a field k, an *algebra* over it is a k-vector space equipped with a bilinear, unitary and associative product.

Just as with groups we define a representation of an algebra as a morphism $\rho : A \to \mathrm{End}(V)$.

Example 1.4 (Trivial representation)

For any group the map $\rho : G \to \mathrm{GL}(V)$ such that

$$g \mapsto 1$$

is the so called *trivial representation* of G.

Definition 1.5 (Regular representation)**.** For any group G, if we take $V = kG$ we get the so called *regular (left) representation* of G as

$$h \circ \sum_{g \in G} \lambda_g g \mapsto \sum_{g \in G} h\lambda_g g$$

this is equivalent to postulate that G acts on kG as the standard left multiplication $h \circ g := hg$ on elements of the form $g \in kG$, and then on all the other elements we extend it linearly.

Example 1.6 (Representation of a permutation)
For $G = S_n$ and $V = k^n$ the map that permutes the canonical basis is a representation, in other words

$$g \circ e_i := e_{g(i)}$$

is a representation.

Example 1.7 (Determinant)
Given a representation $\rho : G \to \mathrm{GL}(V)$, with V a vector space on k, the map

$$\det \circ \rho : G \to \mathrm{GL}(V) \to k$$

is also a representation (in particular is always 1-dimensional).

§1.1 Morphisms of representations

Definition 1.8 (Morphisms of representations)**.** Given a group G and two representations ρ_V and ρ_W over the same field k, a *morphism* between these is a linear map $T : V \to W$ such that is compatible with both ρ_V and ρ_W; that is, we have the following commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ {\scriptstyle \rho_V(g)}\downarrow & & \downarrow{\scriptstyle \rho_W(g)} \\ V & \xrightarrow[T]{} & W \end{array}$$

for any $g \in G$. Equivalently we must have

$$\rho_W(g) \circ T = T \circ \rho_V(g).$$

These are also called *intertwined operators* or *G-equivariant maps*, or *G-morphisms*... they have a lot of names. Anyway we say that a morphism of representations is an *isomorphism* if it is invertible.

What we are saying essentially is, being that $\rho_V(g)$ and $\rho_W(g)$ are matrices, T commutes with all such matrices in the way given above. The set of all intertwined operators is oftend denoted as

$$\mathrm{Hom}(V, W) \geq \mathrm{Hom}_G(V, W)$$

which is always a non empty vector space (!).
Let's now consider two representations V and W, we can investigate what happens to the action of G when we take $V \oplus W$, as we expected

$$g \circ (v, w) = (g \circ v, g \circ w)$$

and no one is surprised. As usual with vector spaces, given two finite dimensional representations V, W, if we choose basis $\mathcal{B}_V$, $\mathcal{B}_W$ theese induce a morphism of representations from V and W to

$$\rho_V : G \to \mathrm{GL}_n(k)$$

and

$$\rho_W : G \to \mathrm{GL}_m(k)$$

in this way $\rho_{V\oplus W}(g)$ has also a nice matrix representation

$$\rho_{V\oplus W}(g) = \left(\begin{array}{c|c} \rho_V(g) & 0 \\ \hline 0 & \rho_W(g) \end{array}\right).$$

Lemma 1.9 (G-equivariant is equivalent to kG-equivariant)
Given a map between two representations V and W of G, we have that this is G-equivariant iff is kG equivariant, given k the field over which V is defined.

Proof. If T is G-equivariant we have that for any $g \in G$ one has

$$T\rho_V(g) = \rho_W(g)T$$

and so extending these to $\tilde{\rho}_V$ and $\tilde{\rho}_W$ we have that, given a certain

$$\sum_{g\in G} \lambda_g g \in kG$$

by linearity of T and $\tilde{\rho}$ we have

$$T\tilde{\rho}_V\left(\sum_{g\in G}\lambda_g g\right) = \tilde{\rho}_W T\left(\sum_{g\in G}\lambda_g g\right)$$

which means that T is kG-equivariant. On the other hand if T is kG-equivariant since $G \subseteq kG$ we have that T is also G-equivariant, and hence we have our thesis. □

Example 1.10 (Dual representation)
Given a representation V of G, it's dual space V^* has naturally a representation structure

$$\left(\rho_{V^*}(g)f\right)(v) = f\left(\rho_V(g)^{-1}v\right)$$

where the inverse is taken to make sure that it is an homomorphism.An equivalent formulation comes with the following commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{f} & k \\ {\scriptstyle \rho_V(g)}\downarrow & \nearrow_{\rho_{V^*}(g)\circ f} & \\ V & & \end{array}$$

a more compact way to write it is $g \circ f = f \circ g^{-1}$. An even more compact way to see this is through matrices, choosing a basis B and it's dual basis B^* of V^* we see that the actions as matrices satisfy

$$\rho_{V^*}(g) = \left[\rho_V(g)^{-1}\right]^{\intercal}.$$

Definition 1.11 (Tensor product of representations)**.** Given two representations V and W of a group their *tensor product* is the space

$$V \otimes W := \langle\{v \otimes w : v \in V, w \in W\}\rangle$$

equipped with the action

$$\rho_{V\otimes W}(g)(v \otimes w) := \rho_V(g)v \otimes \rho_W(g)w.$$

Remark 1.12 (Canonical identification of tensor products)**.** Given spaces W and V^* there is always a canonical bijection

$$F : W \otimes V^* \hookrightarrow \mathrm{Hom}(V, W)$$
$$\sum c_{wf} w \otimes f \mapsto \sum c_{wf} f(v) w$$

which is equivalent to say that on pure tensors our map is just

$$w \otimes f \mapsto F(w \otimes f)(v) := f(v)w$$

and for all the other elements of $V \otimes W$ we extend it linearly. This is just a basis free and abstract way of saying that, given elements $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$ we always have

$$(\mathbf{a} \otimes \mathbf{b})\,\mathbf{c} := (\mathbf{a}\mathbf{b}^\intercal)\,\mathbf{c} = \mathbf{a}\,(\mathbf{b}^\intercal \mathbf{c}) =: \mathbf{a}\langle \mathbf{b}, \mathbf{c}\rangle$$

This gives us immediately that if V and W representations of G, so is $V^* \otimes W$. Since the canonical injection defined above is equivariant (prove it) we hence have that **$\mathbf{Hom}(V, W)$ is a representation of G**. What is the action of this? We have that

$$(g \circ f)(v) = \rho_W(g)\left(f\left(\rho_V(g^{-1})v\right)\right) = \left(\rho_W(g) \circ f \circ \rho_V(g)^{-1}\right)(v)$$

and so if $V = W$ this is just a conjugation of $f \in \mathrm{End}(V)$!

Remark 1.13 (★:Internal Automorphisms)**.** In a way, if we see a representation of G just as a $H \leq \mathrm{GL}(V) \subseteq \mathrm{End}(V)$ monomorphic to G, it is natural to think of the action of G on $\mathrm{Hom})(V, V) = \mathrm{End}(V)$ for any $h \in H$ and $f \in \mathrm{End}(V)$ as the conjugation:

$$h \circ f = hfh^{-1}$$

in other ways this is just an internal automorphism generated by the elements of the subgroup H of $\mathrm{GL}(V)$.

§1.2 Subrepresentations

Definition 1.14 (Subrepresentation)**.** Given a representation V of an object G, a *subrepresentation* of V is a subspace $U \leq V$ such that is G-stable, that is

$$G \circ U \subseteq U$$

which means that acting with G on U always gets us to elements still in U.

Example 1.15 (Fixed point of permutations)

Given $G = S_n$ with it's permutation representation on $V = k^n$, a subrepresentation of V is

$$U = \langle e_1 + e_2 + \cdots + e_n \rangle$$

and in this case is much more than G invariant: is pointwise fixed by the action of G. A non pointwise fixed example is the space W with Cartesian equation

$$\sum_{i=1}^{n} x_i = 0$$

which, if car$K$ $\not| n$ is not pointwise fixed for every point, $w \notin G \circ w$ for any w.

An operation we haven't yet touched is the quotient of spaces, we want to investigate what happens if we take two representation $V \geq U$ of G and consider V/U. We claim that this is a representation with the action

$$\rho_{V/U}(v+U) := \rho_V(g)v + U$$

and we should check that this is actually well defined and is a homomorphism. What does this mean in matrix terms? Given a basis u_i of U and a completion of it to a basis of V, the matrix of $\rho_V(g)$ in this basis has generally the form

$$\rho_V(g) = \left(\begin{array}{c|c} \rho_U(g) & * \\ \hline 0 & \rho_{V/U}(g) \end{array}\right)$$

one of the big questions of representation theory, as always when we deal with matrices, is wether, given a certain representation V of a group G, we can express the action of G on V as acting through block diagonal matrices. From the formula above we see that if we can find a couple of spaces $U \oplus U'$ that are both G stable, we have achieved our goal.

Example 1.16 (Decomposition of a representation of V)
Taking $G = S_n$ and $V = k^n$, since we have

$$\langle e_1 + e_2 + \cdots + e_n \rangle \oplus \left\langle \left\{ \sum_{i=1}^{n} x_i = 0 : x \in k^n \right\} \right\rangle$$

and these are G-stable we have that this is a decomposition of V as a representation.

Remark 1.17 (Lattice of subrepresentations)**.** If V is a representation of G and $U, W \leq V$ are subrepresentations of G then $U \cap W$ and $U + W$ are subrepresentations of G too.

Definition 1.18 (Fixed points)**.** For every representation V of G we can define the set of *fixed points* of V as

$$V^G := \{v \in V : gv = v \quad \forall g \in G\}$$

which can also be seen as the sum of all the subrepresentations isomorphic to the trivial one

$$V^G = \sum_{G \circ u = \{u\}} \langle u \rangle$$

this implies that V^G has naturally a subrepresentation structure.

Lemma 1.19 (Fixed points of the Hom representation)
For any pair V, W of representations of G we have that

$$\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G.$$

seeing $\mathrm{Hom}(V, W)$ as a representation of G through the tensor product.

Proof. From the definition of $\mathrm{Hom}(V, W)^G$ we have

$$\mathrm{Hom}_G(V, W) = \left\{ f \in \mathrm{Hom}(V, W) : \rho_V(g) \circ f \circ \rho_W(g)^{-1} = f \right\}$$

but since

$$\rho_V(g) \circ f \circ \rho_W(g)^{-1} = f \iff \rho_V(g) \circ f = f \circ \rho_W(g)$$

that's just the definition of G equivariant maps, and so we have the thesis. □

Definition 1.20 (Projector)**.** Given a finite group G and a field k we define the *projector* of G in kG as

$$\tilde{P} := \sum_{g \in G} g \in kG$$

this is always an element of $Z(kG)$. Moreover if $\text{char } k \nmid G$ we can define

$$P := \frac{1}{|G|}\tilde{P}$$

which has the nice property that $P^n = P$ for any $n \in \mathbb{N}$.

Lemma 1.21 (Projection to V^G)
Let V be a representation of a group algebra G, then for any $v \in V$ we have

$$\tilde{\rho}_V(P)v \in V^G$$

where $\tilde{\rho}_V$ is the regular representation on a group algebra kG.

Proof. It follows form the fact that $gP = P = Pg$ for any $g \in G$ that

$$g \circ (P \circ v) = gP \circ v = P \circ v$$

and so this satisfies the definition of V^G. □

This implies that $\tilde{\rho}_V(P) : V \to V^G$ is exactly a projection on V^G.

Lemma 1.22 (Kernel and Image of equivariant maps)
For any G-equivariant maps $T : V \to W$ its kernel and image are always G-stable.

Proof. For the kernel, given a $g \in G$ and $v \in \text{Ker}\,(T)$ we wish to prove that

$$g \circ v \in \text{Ker}\,(T)$$

on the other hand, since T is equivariant we have

$$T(g \circ v) = g \circ T(v) = g \circ 0 = 0$$

which is exactly what we wanted. On the other hand if $w \in \text{Im}\,(T)$ we have that exists a $v \in V$ for which we have

$$Tv = w$$

but hence for any $g \in G$ we get

$$g \circ Tv = T\,(g \circ v) \in \text{Im}\,(T)$$

and we have that also $\text{Im}\,(T)$ is G-stable. □

this lemma tells us once more that V^G is a subrepresentation of V.

Definition 1.23 (Irreducible representation)**.** A representation V of G is *irreducible* if its only subrepresentations are itself and the trivial one.

Definition 1.24 (Completely reducible)**.** A representation V of G is said to be *completely reducible*, or *semisimple* if it can be written as a direct sum of irreducible representations.

Example 1.25 (1-dimensional representations)

Every 1-dimensional representation is irreducible, clearly.

Example 1.26 (A non irreducible representation)

Take C_3 with the field $\mathbb{R}$, then its regular representation $\mathbb{R}C_3$ is not irreducible, since we have the subrepresentation

$$U = \left\{a1 + bg + cg^2 : a + b + c = 0\right\}.$$

We try and think to finite dimensional representations which are not completely reducible, but not many pop up to our minds. This makes us wonder if they even exists, indeed for finite groups we have the strong fact result that is Maschke's Theorem:

Theorem 1.27 (Maschke)

Let G be a finite group such that $\operatorname{char} k \nmid |G|$. Let V be a representation of G. Then any subrepresentation of G has a G-stable complement, so as a consequence every finite dimensional representation of G is completely reducible.

Proof. Let $U \leq V$ be G-stable. If W' is such that

$$V = U \oplus W'$$

let $\pi_U^{W'} =: \pi \in \operatorname{Hom}(V, U)$ be the projection onto U along W'. We know that G acts on $\operatorname{Hom}(V, U)$ let's take the projector P of G, then we have that

$$\operatorname{Hom}(V, U)^G = \operatorname{Hom}_G(V, U)$$

and so composing $P \circ \pi$ we get a G-equivariant map. By definition of P we have

$$P \circ \pi = \frac{1}{|G|} \sum_{g \in G} g \circ \pi = \frac{1}{|G|} \sum_{g \in G} \rho_U(g) \circ \pi \circ \rho_V(g)^{-1}$$

where at the RHS the $\circ$ represents function composition and on the LHS represents action. How does this function behave on U? If $u \in U$ we have

$$(P \circ \pi)u = \frac{1}{|G|} \sum_{g \in G} \rho_U(g) \circ \pi \circ \rho_V(g)^{-1}(u) = \frac{1}{|G|} \sum_{g \in G} g^{-1} \circ g \circ u = u$$

so $P \circ \pi_{|U} = \mathrm{id}_{|U}$ but then since $P \circ \pi$ is equivariant (is the composition of equivariant maps), it's kernel must be G-stable, since the kernel is just W', we are done. □

Corollary 1.28 (Strong Maschke)

For any representation V of G we have that the following are equivalent:

- Every subrepresentation of V has a stable complement.
- For any representation V we have irreducible subrepresentations V_i such that

$$V = \bigoplus_{i=1}^{k} V_i.$$

- For any representation V we have irreducible subrepresentations V_i such that

$$V := \sum_{i=1}^{k} V_i.$$

In particular Maschke's theorem tells us that the first condition is true, and so all the other ones must follow as well.

Another strong result about irreducible representations is Schur's lemma:

Lemma 1.29 (Schur)

Le V, W be irreducible representations then

$$\mathrm{Hom}_G(V,W) = \begin{cases} 0 & V \neq W \\ skew - field^a & V \cong W \end{cases}$$

moreover if k is algebraically closed then $\mathrm{End}_G(V) = k\mathrm{id}_V$.

[a] Corpo.

Proof. [1] For the first part we simply have that iff there exists a nonzero $f \in \mathrm{Hom}_G(V,W)$ then we have $\mathrm{Ker}\,(f) \leq V$ is a subrepresentation of V, which means that $\mathrm{Ker}\,(f) = 0$. Thus we have f injective, which means that

$$V \cong \mathrm{Im}\,(f) \leq W$$

which by irreducibility implies $V \cong W$. For the second part we have that, given a $f \in \mathrm{End}_G(V)$ for the algebraic closeness of k we always have an eigenvalue of f, let's call it λ. But this means that

$$0 \neq \mathrm{Ker}\,(f - \lambda I_V)$$

is G-stable, so it must be V. □

Corollary 1.30 (Triviality of action on irreducible representations)

If k is algebraically closed then any $z \in Z(kG)$ acts as a scalar on any irreducible representation.

[1] This proof works only for finite dimension, although the result works for general representations as well.

§1.3 Group functionals and Characters

Definition 1.31 (Character)**.** Let V be a representation of G, then the *character* of V is

$$\begin{aligned}\chi_V : G &\to k \\ g &\mapsto \operatorname{tr}(\rho_V(g))\end{aligned}$$

which is invariant up to G-equivariant maps. This can be naturally extended to kG, and is commonly denoted ad $\tilde{\chi}_V$.

Remark 1.32. For any equivariant map $T : V \to W$ we have

$$\rho_W(g) = T\rho_V(g)T^{-1}$$

and so $\chi_V(g) = \chi_W(g)$ for any $g \in G$.

Remark 1.33. The character is always a class function.

Proposition 1.34 (Properties of the character)
For any V, W representations of G we have:

1. The character is a class function
2. $\chi_{V\oplus W} = \chi_V + \chi_W$
3. $\chi_{V\otimes W} = \chi_V\chi_W$.

Let's now specialize to the case where $k = \mathbb{C}$, if we have that $|G| < \infty$ this means that for $M := \exp G$ we have

$$g^M = 1$$

and so it's associated action is $\rho_V(g)$ on an irreducible representation is a scalar, in particular is an M-th root of unity. Since for any matrix $\rho(g)$ we have that

$$\operatorname{tr}(\rho_V(g)) = \chi_V(g) = \dim V \iff \rho_V(g) = I$$

$$\operatorname{Ker}(\rho_V) = \{g \in G : \chi_V(g) = \dim V\}$$

Definition 1.35 (Inner product on group functionals)**.** On $F(G, \mathbb{C})$ and $F_C(G, \mathbb{C})$ we define an inner product

$$(f, f') := \frac{1}{|G|}\sum_{g\in G} f(g)\overline{f'(g)} = \overline{(f', f)}.$$

Proposition 1.36 (First Orthogonality relation)
If $k = \mathbb{C}$ and $|G| < \infty$, given two representations V, W we have that

$$(\chi_V, \chi_W)_G = \dim \operatorname{Hom}_G(V, W) = \overline{\dim \operatorname{Hom}_G(V, W)} = (\chi_W, \chi_V)_G = \dim \operatorname{Hom}_G(W, V).$$

Proof. We want to compute

$$(\chi_V, \chi_W)_G = \frac{1}{|G|} \sum_{g \in G} \chi_V(g)\overline{\chi_W(g)}$$

since we are on $\mathbb{C}$ the character of the dual representation is just the conjugate, and so we have

$$\cdots = \frac{1}{|G|} \sum_{g \in G} \chi_{V \otimes V^*}(g)$$

so through linearity we get

$$\doteq \chi_{V \otimes W^*} \left(\frac{1}{|G|} \sum_{g \in G} g \right) = \chi_{V \otimes W^*}(P) = \operatorname{tr}\left(P_{|V \otimes W^*}\right)$$

but the trace of a projection is just the dimension of the image, so since P is a projection from $\operatorname{Hom}(V, W)$ to $\operatorname{Hom}_G(V, W)$ we have our result. □

Corollary 1.37 (Ortonormality of characters)

The characters of a set of irreducible representations of a given representation V is an orthonormal set.

Proof. Schur's lemma. □

So putting together the result of Maschke and orthogonality of characthers we get that the set $\operatorname{Irr}(G)$ of irreducible representations of G are always

$$|\operatorname{Irr}(G)| \leq \dim F_C(G, \mathbb{C})$$

and so we have that any χ_V is a $\mathbb{Z}_{\geq 0}$-linear combination of χ_{V_i}.

Corollary 1.38

If $\{V_i, i \in I\}$ is a set of isomorphsim classes of irreducible representations of G we have that any V representation of G is

$$V = \bigoplus_{i=1}^{k} V_i^{\oplus n_i}$$

moreover we have

$$(\chi_V, \chi_i)_G = n_i$$

so the n_i are the fourier coefficients of χ_V in the orthonormal basis χ_i. This means in particular that the multiplicities n_i are unique.